



Evaluating the impact of artificial intelligence in customer service for tech companies

Submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Application (MCA)

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Bonafide Certificate

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Place: Chandigarh

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Declaration

I, Devinderjeet Singh, hereby solemnly declare that the project report titled "**Evaluating the impact of artificial intelligence in customer service for tech companies**" submitted in partial fulfillment of the requirements for the award of the degree of **Master of Computer Application (MCA)** is my original work.

I further declare that:

- This project has been carried out by me during the academic year 2026 under the supervision of *Vikas Kumar Atray*.
- The work has not been submitted previously to any other university, institution, or examination body for the award of any degree, diploma, or certification.
- All sources of information used in this report have been duly acknowledged and referenced in accordance with academic ethics and plagiarism norms.
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Place: Chandigarh

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Abstract

The research focuses on how artificial intelligence (AI) improves customer service in technology companies. Customer service has faced several challenges, like slow responses, poor efficiency, and high operational costs. To address these issues, many companies have introduced new systems such as chatbots, natural language processing (NLP), and automated response tools.

This study looks at how AI affects response time and user satisfaction. To do this, an AI-powered customer service simulation system was used. It collected various performance metrics to compare with other systems.

The results show a big improvement in response time when AI systems are used. However, some issues still exist, such as the lack of empathy and the inability to handle complex customer requests.

In conclusion, artificial intelligence is important for improving customer service.

Chapter1: Introduction

1.1 Background of the Study

In the past few years, Artificial Intelligence has changed how companies operate, especially in customer service. Normally, customer service representatives would talk to customers to answer questions, handle complaints, and solve problems. But now, with so many customer interactions, it's difficult for companies to manage customer service effectively.

Artificial Intelligence has introduced tools like chatbots, virtual assistants, predictive analytics, and Natural Language Processing. These technologies help companies automate customer interactions and improve the overall service experience. They make responses faster, ensure service is available 24/7, and provide accurate answers every time. Artificial Intelligence is playing a big role in making customer service more efficient. Companies are using it to improve their customer service processes.

Tech companies are increasingly relying on AI for their customer service. They want to manage a large number of customer messages at the same time and do it efficiently. These AI tools can operate around the clock, analyze a lot of data, and offer customers suggestions based on their preferences. Switching from traditional systems to AI has made a significant difference. Companies can achieve more with fewer resources, and customers are generally more satisfied. This study explores how AI is transforming customer service. We aim to find out if AI leads to happier customers and if it helps companies respond more quickly.

The project has two main parts:

- The first part is creating an AI dashboard.
- It helps users view customer service data.

Users can see how AI influences customer experiences, both positively and negatively. The goal is to understand the impact of AI on customer service performance. We focus on response time, customer satisfaction, and overall experience.

1.2 Problem Statement

Despite advances in technology, traditional customer service is still facing some issues with the new technology. The main problems are that people aren't getting quick answers, the service isn't always reliable, and it's expensive to keep running. The system can't handle many customers at once. Human customer service agents aren't always available to help, and they don't always do a good job because they can get overwhelmed or lack experience. More and more people are using customer service, and they expect fast responses. Traditional systems aren't up to the task, which makes customers unhappy and harms the company's reputation. Customer service is important, but the old systems aren't working well. These systems need to handle lots of customers and provide good service.

Even though AI systems provide solutions, we still need to check if they really work well in everyday situations. Important questions like whether AI actually makes customers happier, how it affects response time, and if it can completely replace talking to a human are still important to study.

This project is about checking how well AI performs in customer service by looking at numbers and comparing them with the old methods. It also aims to find out what AI is good at and what it is not so good at.

1.3 Objectives and Scope of the Project

a) Objectives of the Study

The main aim of this project is to look at how Artificial Intelligence is affecting customer service in technology companies. The objectives are as follows:

- Analyze customer service data using an AI-driven dashboard.
- Track key performance indicators like customer satisfaction and how quickly responses are given.
- Explore the link between response time and customer satisfaction.
- Apply machine learning methods to predict customer satisfaction levels.
- Use natural language processing to analyze the feelings expressed in customer reviews.
- Compare traditional customer service approaches with those that use AI.
- Provide useful suggestions for improving customer service through the use of AI.

The project also aims to show how AI techniques can be applied in real situations by using tools like Streamlit and Python.

b) Scope of the Project

The project is about creating a system that helps companies understand how well their customer service is performing using intelligence. This system is a website that allows users to view customer service data. It uses a tool called Streamlit, which can store, analyze, and even interpret what people are saying.

The project focuses on parts of customer service. It allows users to look at customer service data stored in a database that is organized and easy to manage. This database is called SQLite. It helps keep all the information safe and easy to use. The system also lets users add customer information easily. This means the system is not only for viewing information but also for taking action with it. The project is about improving customer service using artificial intelligence.

The dashboard is really useful because it lets users view data in a way that makes sense to them. They can choose which type of data they want to see depending on how the artificial intelligence is being used, such as if it's used a lot, a little, or not very much. This helps

people examine the data in more detail. The dashboard also identifies and shows important information like average customer satisfaction, how long it takes to get a response, and the total number of records. It uses pictures and graphs to show what's happening with the data, like what's normal and what's different. It also uses special kinds of graphs to show how different things are connected, such as graphs that show how data is spread out and graphs that show how things are related to each other. This is helpful for understanding what is really going on.

The dashboard is really helpful because it allows users to view data in a way that makes sense to them. They can choose what kind of data they want to see depending on how the artificial intelligence is being used, such as if it's being used a lot, a little, or not much at all. This helps people examine the data in detail. The dashboard also identifies and shows important information like the average customer satisfaction, how long it takes to get a response, and the total number of records. It uses pictures and graphs to display what's happening with the data, like what is normal and what isn't. It also uses special kinds of graphs to show how different factors are connected, such as graphs that show how things are spread out and graphs that show how things are related to each other. This is helpful for understanding what is really happening.

The company uses a machine learning part that applies Linear Regression to determine how satisfied customers are with how quickly they get a response. This helps show how the way the company operates influences what customers think about the service.

The system also includes a Natural Language Processing section that uses TextBlob. This part checks what people write in their feedback to figure out if it's positive, negative, or neutral. It does this by analyzing the words used. The Natural Language Processing part is important because it helps us better understand what customers think about the service by looking at their feedback.

- The machine learning part uses Linear Regression to predict how satisfied customers are.
- The Natural Language Processing part uses TextBlob to analyze the sentiment of what customers write.

This helps us understand how response time and the Natural Language Processing part together impact customer satisfaction and the overall user experience. Customer satisfaction is very important, and the Natural Language Processing part helps us understand it better.

The project also includes a feature for generating reports. This allows users to download a summary of the insights gained. They can use this for documentation purposes or to make informed decisions.

The project has certain limitations. It only uses machine learning models and works with sample data. It doesn't include deep learning models or real-time connections to company

systems. Despite these limitations, the project still provides a good example. It helps in evaluating customer service systems that use AI.

1.4 Overview of Customer Service Systems

a) Overview of Existing Customer Service Systems

Customer service is usually handled by people. It's done manually, which means customer service representatives talk to customers on the phone, send emails, or chat with them on the website. These representatives are the ones who receive questions from customers. They try to understand the situation and figure out the best way to resolve it. This method has its advantages because it allows customer service representatives to communicate directly with customers and show empathy by understanding their feelings. However, there are some challenges with this approach. One major issue is that customer service is not always available. Most customer service teams operate during the day and are not available at night or on weekends. This can make it difficult for customers to get help when they need it most. Additionally, during busy times when many customers are contacting support at once, it can take a long time to get a response.

Another issue with customer service is that it can vary a lot depending on who you speak with. For example, different people might give you different answers to the same question. This happens because people can make mistakes, get tired, or have different levels of skill. All of these factors can affect how well they perform their job.

When many people are asking for help at the same time, it can be really tough to keep up. Companies need to hire people to handle these questions, which costs a lot of money. It's also hard to manage all the conversations happening at once.

Customer service systems also don't make good use of the information they collect. They might have a lot of data, but they don't use it to understand what's happening or to improve things. They can't find patterns or predict what might happen next.

These problems show that we need better ways to handle customer service. That's why many companies are starting to use smart customer service systems, like AI-based ones. AI-based systems can help fix these issues and improve the experience for customers.

b) Overview of AI-Based Customer Service Systems

Customer service systems that use intelligence are a new way to handle customer interactions. They use things like machine learning and data analysis to function. These systems are designed to answer questions on their own, observe what customers do, and improve the service over time.

These systems are different from traditional methods. They can work 24/7, so customers can get help whenever they need it. Computer programs that talk to customers, like

chatbots, can handle many questions at once. This means that customer support workers don't have to do as much work, and customers get faster responses.

The system we built for this project has features similar to other AI-based customer service systems. It uses a database called SQLite to store and manage customer information. We also used something called Streamlit to create an interface. This interface allows users to view and analyze data over time. The customer service system we made uses intelligence to operate, and artificial intelligence is used to make it work.

Data analytics is very important for Artificial Intelligence systems. The dashboard uses pictures like bar charts and graphs to show how different things are connected, such as how long it takes to respond and how happy the customers are.

They use something called Machine Learning to predict how happy the customers might be based on how long it takes to respond. This helps companies see how improving the system can make customers more satisfied.

The project also uses Natural Language Processing and a tool called TextBlob to analyze the words people use. This helps the system understand how customers are feeling, so we can learn more about what they like and what they don't like.

1.5 Technologies Used

The project is implemented using various modern technologies and tools:

- ❖ Python: The primary programming language used for development
- ❖ Streamlit: Used to build an interactive web-based dashboard
- ❖ Pandas: Used for data manipulation and analysis
- ❖ Matplotlib: Used for data visualization and plotting graphs
- ❖ Scikit-learn: Used to implement machine learning models (Linear Regression)
- ❖ TextBlob: Used for Natural Language Processing and sentiment analysis

These technologies work together to build an AI-powered customer service analysis system. By combining data visualization, machine learning, and natural language processing, they create a platform. This platform helps assess how AI affects customer service. It uses data visualization to present key insights. Machine learning and natural language processing are essential for analyzing customer data. The AI system offers a clear view of how things are going. It helps make customer service better. The system uses AI to examine data and clearly shows the impact of AI.

Chapter 2: System Study

2.1 Introduction to Artificial Intelligence

Artificial Intelligence is a method that allows machines to think and act like humans. It's a big part of the computer world and is used in many different places, such as hospitals, banks, schools, and when you call a company for assistance.

These machines are designed to perform tasks that people normally do, like understanding what we say, finding patterns, solving problems, and predicting what might happen. They use various tools, such as machine learning, deep learning, and natural language processing.

Machine Learning is a part of Artificial Intelligence that enables machines to learn and make decisions on their own. It works by analyzing data, identifying patterns, and making accurate predictions. For example, we used a Linear Regression model to determine how satisfied customers are when they receive help quickly. This demonstrates how machine learning can be applied in a practical and effective manner.

Natural Language Processing is part of Artificial Intelligence that helps computers understand how people speak and write. It lets machines figure out what people mean when they talk or write. In this project, we use Natural Language Processing to check what customers say and how they feel about things – whether it's good, bad or just okay.

Types of Artificial Intelligence

Artificial Intelligence can be divided into different types depending on what it can do and how it functions.

1. Narrow Artificial Intelligence

Narrow Artificial Intelligence is designed to perform a specific task very well. Examples include chatbots, recommendation systems, and virtual assistants. Our current project uses Narrow Artificial Intelligence to understand customer satisfaction and gather their opinions.

2. General Artificial Intelligence

General Artificial Intelligence refers to systems that can perform tasks that humans can do. These systems are still in development and are not yet available.

3. Reactive Machines

Reactive machines are straightforward. They perform actions only when given specific instructions. They don't keep track of previous interactions. A simple chatbot is an example of a reactive machine.

4. Limited Memory AI

Limited Memory AI systems are a bit more advanced. They use past experiences to help make decisions for the future. Many machine learning models fall into this category.

AI systems are great at looking at a lot of information and getting better at what they do the more they learn. This makes them perfect for jobs like handling many customer calls and giving responses.

The change came when AI started being used for customer service. Now companies can interact with customers in new ways. AI can take over tasks and help businesses understand their customers better, which means they can work more efficiently, save money and make customers happier.

2.2 AI in Customer Service – Concept and Evolution

Customer service has changed a lot over the years. In the past, people used to talk face to face to get help. Now, computers and machines play a big role in providing customer service.

At first, customer service was all done by people. They would answer calls, reply to emails, or help customers in person.

As technology improved, companies started using tools like email and live chat on their websites. This made it easier for customers to get help, but people were still needed to answer their questions.

When artificial intelligence came into use, it made a big difference for customer service. Now, machines can answer many questions on their own. AI can understand what customers are asking and give them quick answers. It uses tools to figure out what customers mean and provide the right information. Customer service is still a mix of people and AI, and AI has become a major part of it.

Customer service systems that use artificial intelligence are really helpful. These systems let customers get assistance whenever they need it, without having to wait for a person to respond.

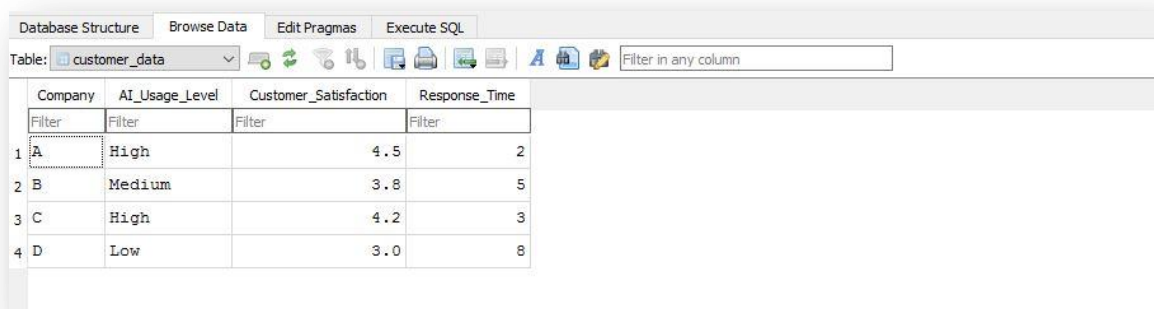
Other great things about customer service systems that use artificial intelligence include:

- They can answer questions quickly.
- They always provide the same answers so customers receive the same information every time.
- They can handle many questions at the same time.
- They can listen to what customers are saying and use that information to make the service better.

Customer service systems that use artificial intelligence are very useful because they can do all these things.

They are available twenty-four hours a day and seven days a week. This is great for customers who need help at any time. Customer service systems that use artificial intelligence can help customers quickly.

In this project, we created a computer program called a dashboard to evaluate the performance of customer service. The project uses an SQLite database named ai_data.db to keep track of customer service records. This database holds a lot of details, such as the company name, the extent to which AI is used, customer satisfaction scores, and response times. We use the data from ai_data.db to create visualizations, make predictions, and analyze people's feelings and opinions through the dashboard.



	Company	AI_Usage_Level	Customer_Satisfaction	Response_Time
	Filter	Filter	Filter	Filter
1	A	High	4.5	2
2	B	Medium	3.8	5
3	C	High	4.2	3
4	D	Low	3.0	8

Figure 2.1: SQLite Database Table – customer_data

The program allows users to:

- View customer satisfaction levels and response times
- Use visuals and graphs to better understand the data
- Predict customer satisfaction using a special type of computer learning
- Analyze comments to determine customer sentiment

By using computer learning in this program, it demonstrates how new customer service systems can leverage computers, data, and smart ideas to make better decisions.

The customer service dashboard is a tool that helps us track how well customer service is performing. Customer service performance is important for companies to know. The dashboard is used to check how customer service is doing.

There are some issues with using computer learning in customer service. Sometimes it gets confused with questions it doesn't understand, and it might give incorrect answers. Because of this, many companies use a mix of computer learning and human support. The customer service dashboard is an example of how customer service performance can be measured using computer learning along with other tools.

Applications of Artificial Intelligence in Customer Service. Artificial Intelligence is widely used in customer service across various fields. Some of the ways it is applied include:

- Chatbots: These are used to answer customer questions automatically.
- Assistants: They offer customer support that you can interact with through conversation.
- Automated Email Responses: Artificial Intelligence can send replies to customer emails.
- Sentiment Analysis: This helps companies understand what customers are feeling or thinking.
- Customer Behavior Analysis: Artificial Intelligence identifies patterns in how customers engage with companies.
- Recommendation Systems: These suggest products or services that customers might be interested in based on their preferences.

In this project, we use sentiment analysis and customer satisfaction prediction to evaluate how effective customer service is. We use Artificial Intelligence to carry out these tasks. The aim is to use Artificial Intelligence to gain a better understanding of customers and improve customer service.

2.3 Review of Previous Studies

Many people who do research and organizations have studied how artificial intelligence affects the way we help customers. They found that artificial intelligence has both good and not-so-good sides.

In 2018, Davenport and Ronanki did a study about how companies use intelligence to improve their work, including helping customers. They found that artificial intelligence is good at doing repeated tasks and making things more efficient, but we still need people to make big decisions.

In 2021, Huang and Rust looked at how artificial intelligence is used in jobs that involve helping people. They believed that artificial intelligence can make customers happier by providing more accurate answers. They also said that it's important for people to talk to each other so they can trust each other and feel good about it.

In 2020, Accenture's research showed that Artificial Intelligence can reduce customer service costs by up to 30 percent. At that time, AI was also found to help make customers more satisfied with the service they receive. The researchers emphasized that it's important to combine AI with existing systems to achieve the best outcomes.

Looking at the use of chatbots, customers tend to appreciate fast responses. However, they can get frustrated when chatbots don't understand their questions. This means that the developers of Natural Language Processing models need to keep improving them. AI and

NLP models play a big role in customer service, and ongoing improvements are essential to keep customers happy.

The project we're currently working on is built on some new findings. It uses a system that combines a few different technologies and features. Basically, this project is all about the project itself and what it can achieve for us.

Here are the details:

- ❖ **Data Analysis (Pandas):** We use Pandas to load and organize the customer service data from the SQLite database. This data is used by the project. The project uses the data to help us. Pandas helps us look at the data, filter it, and analyze it. With the help of the project and Pandas, we can calculate things like customer satisfaction with the project and customer service on average, and how long it takes the project to respond to customers.
- ❖ **Visualization (Matplotlib):** We use Matplotlib to create graphs and charts that show what the data means for the project. These include bar charts, histograms, and scatter plots. The project uses these graphs to help people see patterns and trends in the data, such as how using intelligence affects response time and customer satisfaction. The project is important. The project uses the data to improve itself.
- ❖ **Machine Learning (Linear Regression):** We use a Linear Regression model to predict how satisfied customers are based on how quickly we respond. This helps us understand how efficient our service is and how that affects customer satisfaction, making it easier to make decisions using data.
- ❖ **Natural Language Processing – NLP (TextBlob):** We use TextBlob to analyze what customers say in their feedback. It looks at the text they write and determines if they are happy, unhappy, or neutral. This helps organizations like ours understand what customers truly think and feel.

The project shows how these technologies can be combined in one system to assess how well customer service is performing. The project is about creating a platform that evaluates customer service performance using these technologies. The goal of the project is to evaluate customer service performance through the use of these technologies.

2.4 Research Gaps Identified

Even though there's been a lot of research on using Artificial Intelligence in customer service, there are still some areas that aren't well covered. One major issue is that there aren't many real-world systems that clearly show how AI works in practice. A lot of studies look at models or solutions used by big companies, but those aren't practical for students or small businesses. As a result, AI techniques aren't being used in a complete or practical way.

Another problem is that most research looks at just one or two parts of AI, like machine learning or Natural Language Processing. Not many projects bring together machine

learning, NLP, data visualization, and database management. For AI systems to be really useful, they need to include more different parts and work together better.

There's also a lack of tools that let regular users view customer service details. Many of these systems are difficult to use because they require a lot of technical knowledge. Also, even though AI helps make things more efficient, we still need to understand how it impacts customer satisfaction in a way that can be measured. We need to look at customer service data to see how it affects customer satisfaction. We should check how quickly responses are made and how that influences customer opinions. We need to analyze customer service data to get a clearer picture.

The project I am working on fills the gaps by:

- Creating a complete system from scratch using Streamlit.
- Bringing together a database, analytics, machine learning, and natural language processing all in one place.

The database used in this project is SQLite.

- Building a dashboard to simplify data analysis.
- Including a feature for real-time predictions and sentiment analysis.
- Allowing users to add, filter, and analyze data anytime.

This project is straightforward and user-friendly, making it great for learning and for use in various projects.

One key area that needs more attention is the lack of interactive dashboards. These dashboards should include visualization, prediction, and sentiment analysis. However, most current systems focus only on analytics or automation, and they don't include all these features together.

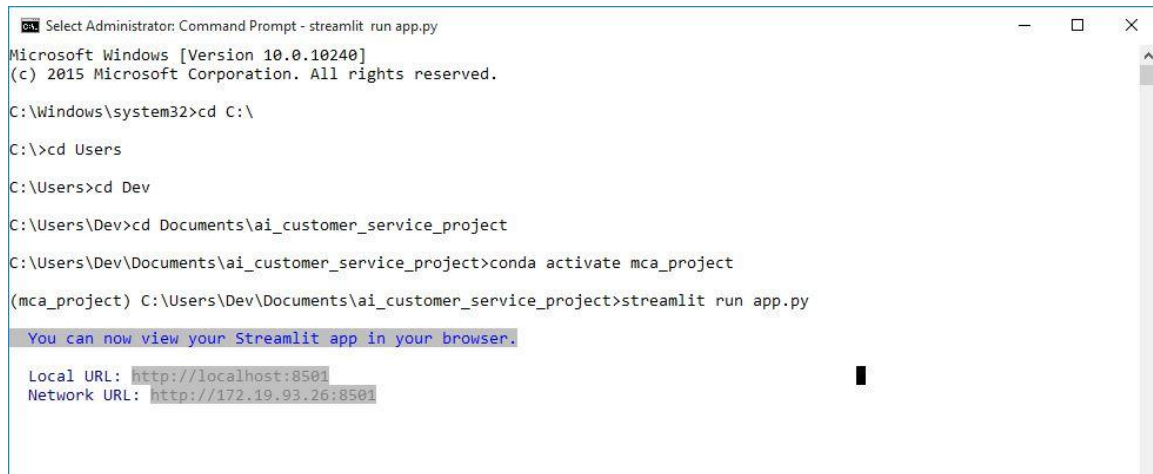
This project addresses that issue by creating a single application that combines visualization, prediction, and sentiment analysis. The app is built using Streamlit and SQLite.

Application Execution Using Streamlit

We use Streamlit to run our system, and we do this through the command prompt. Streamlit helps us easily convert our dashboard into a web application. To start the application, we use the following command:

```
streamlit run app.py
```

When we do this, Streamlit creates a URL. This URL allows users to view the AI Customer Service Dashboard in a web browser. So we can see that our system is actually working and that it has been successfully deployed.



```
Select Administrator: Command Prompt - streamlit run app.py
Microsoft Windows [Version 10.0.10240]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd C:\
C:\>cd Users
C:\Users>cd Dev
C:\Users\Dev>cd Documents\ai_customer_service_project
C:\Users\Dev\Documents\ai_customer_service_project>conda activate mca_project
(mca_project) C:\Users\Dev\Documents\ai_customer_service_project>streamlit run app.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.19.93.26:8501
```

Figure 2.2: Running the AI Customer Service Dashboard Using Streamlit

Chapter 3: System Analysis

System analysis is part of creating software. It involves looking at the current system, finding out what isn't working, and then developing solutions. The main purpose of system analysis is to understand how information flows through the system, how people interact with it, and how technology can be used to improve things and make decisions easier.

When it comes to helping customers, companies often face many issues. They take a long time to respond to questions, don't always treat customers the same way, don't listen well to what customers have to say, and it's expensive to handle things this way. The old method of customer support isn't effective anymore. Companies are receiving a large number of customer requests and can't keep up. Because of this, they are starting to use AI-powered systems to automate tasks and improve customer satisfaction. They want to use artificial intelligence to enhance customer service.

The project we're suggesting, "Evaluating How Artificial Intelligence Affects Customer Service in Tech Companies," looks into how customer service is handled when AI tools like Machine Learning and Natural Language Processing are used. The system combines database management, data visualization, predictive analytics, and sentiment analysis into a single dashboard.

This section explores the system's limitations, discusses the proposed AI-based solution, and explains the feasibility of the solution. It also outlines how the system is designed to function.

3.1 Current System Analysis

Many companies still use outdated customer service systems. These systems mostly depend on people to handle customer calls, reply to emails, and chat online.

When dealing with customer service, people often manage complaints and questions one by one. They use spreadsheets or basic databases to keep records. While this helps track information, it's not enough when there are many customers to handle at once.

Most customer service systems only store customer data without really understanding it. They collect feedback through surveys or forms, but organizations usually don't get useful insights from it. This makes it hard for them to improve their services or know what customers truly need. Customer service systems should be better at understanding both feedback and customer behavior to truly support businesses. Customer service is essential, and these systems need improvement to better serve both customers and organizations.

Customer service can sometimes be really slow. When someone calls with a problem, the person on the phone has to take time to understand what's going on and figure out a solution. This process can take a while, especially when there are a lot of calls coming in at

the same time. Customers get frustrated waiting for help, and that makes them unhappy with the service.

Old systems aren't very good at predicting what customers might do next. They can't look at past actions to guess what might happen in the future. Because of this, companies can't fix problems before they even occur.

Also, many systems don't provide real-time information. They usually send reports at certain times, which means people have to wait to get the information they need to make decisions. This makes it difficult for companies to act quickly when customers need something to change.

The system we're using now isn't very effective at bringing everything together. We store customer information, create reports, and examine the data through various programs. This makes things more complicated and takes longer to complete.

The new project we're discussing will improve this by combining customer data, images of the data, machine learning, and natural language processing into one dashboard. This way, customer data management will be centralized, with visualization, machine learning, and NLP all in one place.

3.2 Existing System Workflow

The way traditional customer service systems work usually goes like this:

- 1) A customer sends in a question through the phone, email or chat.
- 2) Human agents get the question. Look at it.
- 3) The customers information is written down in records by hand.
- 4) Agents answer the customers questions.
- 5) We ask the customer what they think after we are done helping them.
- 6) Reports are made by hand for the people in charge to look at.

This way of doing things takes a lot of work and a lot of time. It also means mistakes can happen because people are doing everything. Customer service systems are, like this. That is why customer service systems can be slow. Customer service systems need to be improved because customer service systems are important.

3.3 Limitations of Existing Systems

The customer service systems that are already in place have some problems that make them work well and make customers less happy.

3.3.1 Manual Data Handling

A lot of companies still use ways to write down and look at what customers are saying. When people have to enter data by hand it can lead to mistakes doing the thing twice and things not matching up.

3.3.2 Lack of Real-Time Insights

The systems do not give us information, about what is happening right now. Managers usually get reports a time after something has happened which makes it hard to make decisions quickly.

3.3.3 Limited Scalability

When customer service systems have to deal with more customers they have a hard time handling all the requests. The customer service systems get overwhelmed. Companies need to hire more people to help with the customer service systems.

3.3.4 No Predictive Analytics

We don't have systems that can tell us if our customers are happy or what they might need in the future. This is because these systems don't use machine learning models. As a result, they can't predict customer satisfaction or future service trends.

3.3.5 Poor Feedback Analysis

When customers provide feedback, it's usually reviewed by a person. This makes it difficult to fully understand how our customers feel or whether they are satisfied with our service. Customer feedback is something we need to pay attention to.

3.3.6 Fragmented Architecture

Our system has different parts that handle various tasks, such as storing data, analyzing data, and generating reports. These parts are not integrated well, which causes issues. It makes our processes less efficient. We often end up doing the same task more than once.

3.3.7 High Operational Cost

When we use people to assist our customers, it costs a lot. We have to pay for staff and use more resources, which makes our customer support system more expensive to run. Customer support systems that rely on people are costly.

3.3.8 Lack of Intelligent Automation

Our old systems can't do things automatically, like answering questions or looking at data to find patterns. These are tasks we have to do manually. We end up answering the same questions repeatedly.

These issues show that we need a system that uses intelligence to help with customer support. We need a system that can do things on its own and assist our customers. Customer support is something we need to improve.

3.4 Proposed System Overview

To improve existing systems, this project is creating a tool named the Customer Service Dashboard that uses Artificial Intelligence. The Customer Service Dashboard is developed using Streamlit and SQLite.

This new system combines various technologies into a single platform, allowing companies to assess the performance of their customer service.

The Customer Service Dashboard performs several functions:

- It stores customer information using SQLite
- It features an interactive dashboard built with Streamlit
- It displays visuals for customer service performance metrics
- It uses machine learning to make predictions
- It applies natural language processing for analyzing customer sentiments
- It allows users to filter data dynamically
- It creates reports based on the data

The dashboard is a tool that allows users to view customer happiness, response time, and how the artificial intelligence system is being used, and it does all of this in real time. It provides a way for people to see customer satisfaction levels and the patterns of artificial intelligence usage as they happen.

3.5 Objectives of the Proposed System

The main objectives of the proposed system are:

- It aims to make it simpler to access customer service information.
- It aims to display how customer service is performing through images and charts.
- It aims to predict customer satisfaction using machine learning.
- It aims to analyze what customers are saying about the service using computer techniques.
- It aims to provide a screen where users can view key information.
- It aims to reduce the time spent creating reports manually.
- It aims to support decision-making by providing up-to-date information on customer service.

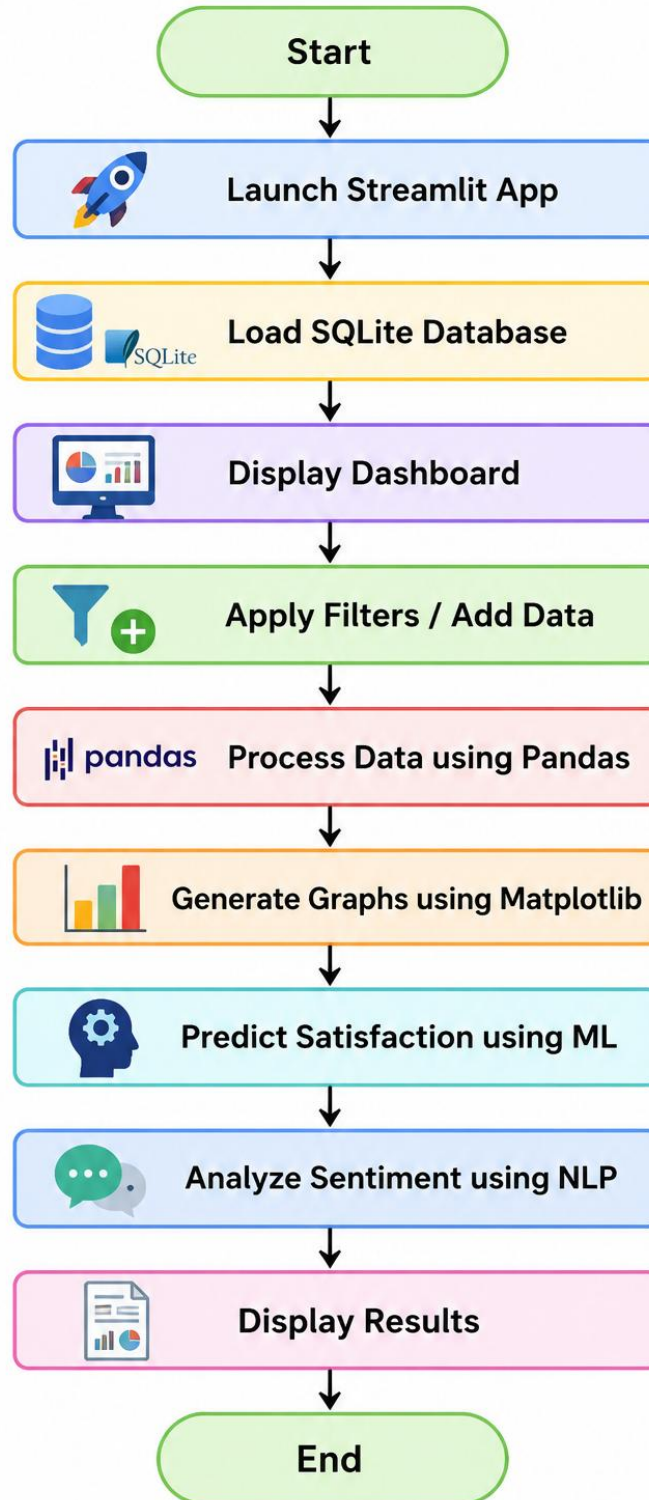


Figure 3.1: Workflow Diagram of AI Customer Service Dashboard

3.6 Modules of the Proposed System

The proposed system consists of several modules.

3.6.1 Database Module

This part is responsible for storing customer service data. It uses a SQLite database named ai_data.db.

The database includes the following fields:

- Company
- AI Usage Level
- Customer Satisfaction
- Response Time

This database helps in storing and retrieving customer service data efficiently.

3.6.2 Dashboard Module

The dashboard module is built using Streamlit. It allows users to interact with the system in a simple and easy way.

Some of the main features include:

- Displaying important metrics.
- Letting users view datasets.
- Enabling data filtering.
- Assisting with the visualization of trends through the dashboard module.

3.6.3 Data Visualization Module

We use Matplotlib to make graphs and charts like

- Bar charts
- Histograms
- Scatter plots

These graphs and charts really help people understand how customer service is doing.

3.6.4 Machine Learning Module

This part uses Linear Regression to guess how happy customers are with how it takes to get a response.

This guessing model helps companies

- See how well their service is working
- Understand what customers are doing
- Make their responses better

3.6.5 NLP Sentiment Analysis Module

We use TextBlob to look at what people're saying in their feedback. The NLP Sentiment Analysis Module puts feelings into groups, like

- I. Positive feelings
- II. Negative feelings
- III. Neutral feelings

This really helps companies understand what customers think about them.

3.7 System Workflow of Proposed System

The following are the steps in our suggested system:

- The application is launched by the user via Streamlit.
- The SQLite database is used to load data.
- The data is then filtered by the user according to the degree of AI usage.
- Metrics and visualizations are displayed on the dashboard.
- Customer satisfaction is predicted by a machine learning model.
- To examine the sentiment of client feedback, we employ an NLP module.
- We produce reports for examination.

This process enables us to assess customer service intelligently.

3.8 Feasibility Study

We do a feasibility study to see if the proposed system is an idea and if it will work.

The feasibility study looks at a things:

- Technical Feasibility
- Economic Feasibility
- Operational Feasibility

3.8.1 Technical Feasibility

The proposed system is technically feasible because it uses open-source technologies. We are using the following technologies, for the proposed system:

Technology	Purpose
Python	Programming Language
Streamlit	Dashboard Development
SQLite	Database Management
Pandas	Data Analysis
Matplotlib	Visualization
Scikit-learn	Machine Learning
TextBlob	NLP Sentiment Analysis

This means the proposed system feasibility study is looking at the Technical Feasibility of the proposed system and we think it is a good idea. The proposed system uses these technologies to make sure it is technically feasible.

3.8.2 Economic Feasibility

From a financial perspective, the project is a concept since it may be developed using free and open-source tools. We are in a better financial position.

- Software licenses are free.
- The expense of maintenance is minimal.
- Not much hardware is required.
- We don't require a large team to do the task.
- The system can generate reports on its own, saving us time.
- Artificial intelligence helps us complete tasks and reduce operating costs.

3.8.3 Operational Feasibility

Because it is straightforward to use and comprehend, the system we are proposing is easy to operate.

There are advantages in terms of operations.

- The user interface is straightforward.
- There is interaction on the dashboard.
- Data management is simple.
- We receive timely insights
- Data is automatically analyzed by the system

The Economic Feasibility and Operational Feasibility systems are still useful for those with limited technological expertise.

3.9 Conclusion

The main issues with customer support systems were examined in this chapter. Older systems rely heavily on human labor. lack sophisticated analytical tools.

We propose an AI-powered customer service dashboard to address these problems. Natural language processing, database administration, machine learning, and data visualization are all integrated into this dashboard. The system's components, how it functions, and whether it is feasible were also covered in the chapter.

The new technology provides IT organizations with an intelligent and adaptable method of evaluating customer service performance.

Chapter 4: System Design

4.1 System Architecture

The components of the AI Customer Service Dashboard are interdependent. It's similar to a team gathering data, analyzing it, attempting to predict future events, and presenting the findings.

Because the system is constructed in layers, each component performs a specific function. In this manner, machine learning, natural language processing, the frontend, and the backend database all function as a single unit.

The system was constructed using Python. What the user sees and engages with is streamlit. All of the customer service data is kept in a SQLite database. Additionally, the system makes use of Pandas to analyze the data, Matplotlib to create charts, Scikit-learn to make predictions, and TextBlob to determine people's emotions.

The architecture of the system contains the following major components:

1. Layer of User Interface

Users can interact with the system with the aid of the User Interface Layer. Streamlit, which creates a web-based dashboard, is used in its construction. This interface enables users to perform tasks such as:

- View customer support information
- Apply data filters
- Update the system with new records
- Use charts and graphs to examine data.
- Estimate the level of client satisfaction
- Examine the tone of client comments
- Get customer service reports

The interface's objective is to make its functions simple and straightforward for consumers to use.

2. Application Processing Layer

Everything that occurs within the system is handled by the Application Processing Layer. It considers what the user desires, does the numbers and makes decisions.

This layer performs various tasks, including:

- Working with data using Pandas
- Using statistics to comprehend the information
- Making sense of the data by filtering it
- Preparing the data for visualization
- Making predictions about the data using machine learning

The Application Processing Layer is crucial because it facilitates the seamless interaction between the database and the user interface.

3. Database Layer

Every client service record is kept in the database layer. This project makes use of a SQLite database called ai_data.db.

The database keeps:

- Company name
- AI usage level
- Customer satisfaction score
- Response time

Because SQLite is lightweight, user-friendly, and appropriate for small and medium-sized applications, it was chosen.

4. Machine Learning Layer

The Scikit-learn Linear Regression algorithm is used in the Machine Learning Layer. This layer performs a function.

- The prediction model is trained by it.
- Response time is an input for the Machine Learning Layer.
- The customer satisfaction score is predicted by the machine learning layer.
- The association between customer happiness and service speed is analyzed with the aid of the machine learning layer.
- Making decisions is aided by the prediction system.

5. Natural Language Processing Layer

The TextBlob library is used by the NLP layer to examine client feedback. This layer performs the following tasks:

This layer carries out:

- Sentiment analysis
- Analysis of polarity

Classification of feedback into : Positive, Negative and Neutral

Because it helps businesses learn what customers like and dislike as well as how they feel about the company, the sentiment analysis capability is incredibly helpful. The NLP layer and the sentiment analysis feature are crucial because they enable businesses to comprehend the feelings and viewpoints of their customers by analyzing their feedback using the TextBlob library.

6. Visualization Layer

The Visualization Layer uses Matplotlib to create images using graphs and charts.

The system produces:

- Bar graphs
- Histograms
- Plots of scatter

People can really notice patterns and trends in the customer service data thanks to these images. By creating these images, the Visualization Layer does a great job of assisting users in understanding the customer service data. Using The Visualization Layer to create these graphs and charts makes the customer service data more comprehensible.

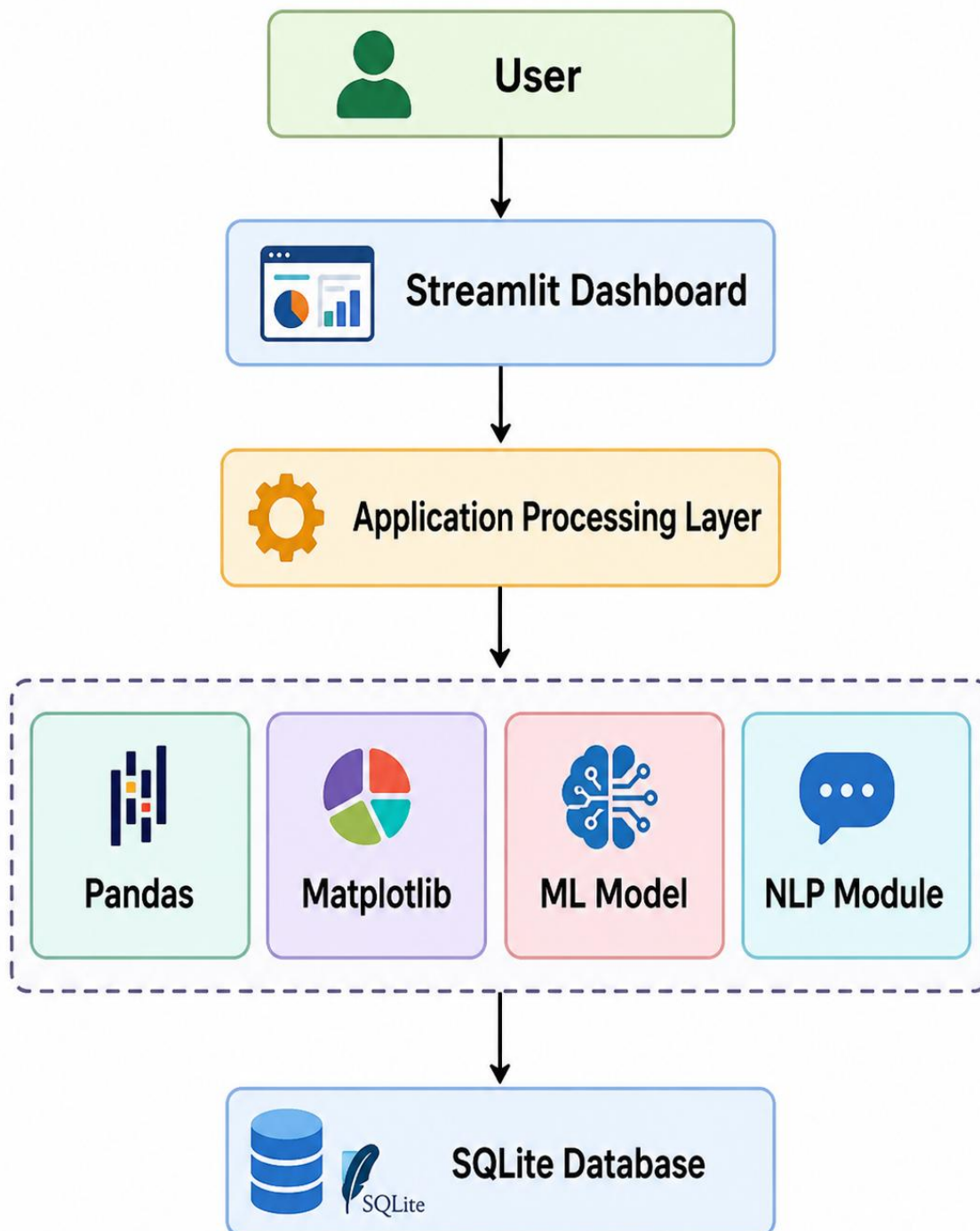


Figure 4.1: System Architecture Diagram

4.2 Data Flow Diagrams (DFD)

Data Flow Diagram or DFD for short is a way to show how information gets from one place to another in a system. It is like drawing a map to see where data comes from how it gets changed, where it gets stored and where it goes.

DFD helps us understand things like:

- How data moves around
- The steps that data goes through to get processed
- How different parts of the system talk to each other
- What outside things, like people or other systems connect with our system

The proposed project includes:

- Level 0 DFD (Context Diagram)
- Level 1 DFD

4.2.1 Level 0 DFD (Context Diagram)

The Level 0 DFD shows our system as one process. In this system:

- ✓ We get customer data and feedback from users.
- ✓ Our system then works on this information.
- ✓ Finally it gives us analysis, graphs, predictions and reports.

Main Inputs

- Customer service data
- What users choose to filter
- What customers write in feedback

Main Outputs

- Analytics, on a dashboard
- Graphs and pictures
- Results of sentiment analysis
- Results of predictions
- Reports that can be downloaded

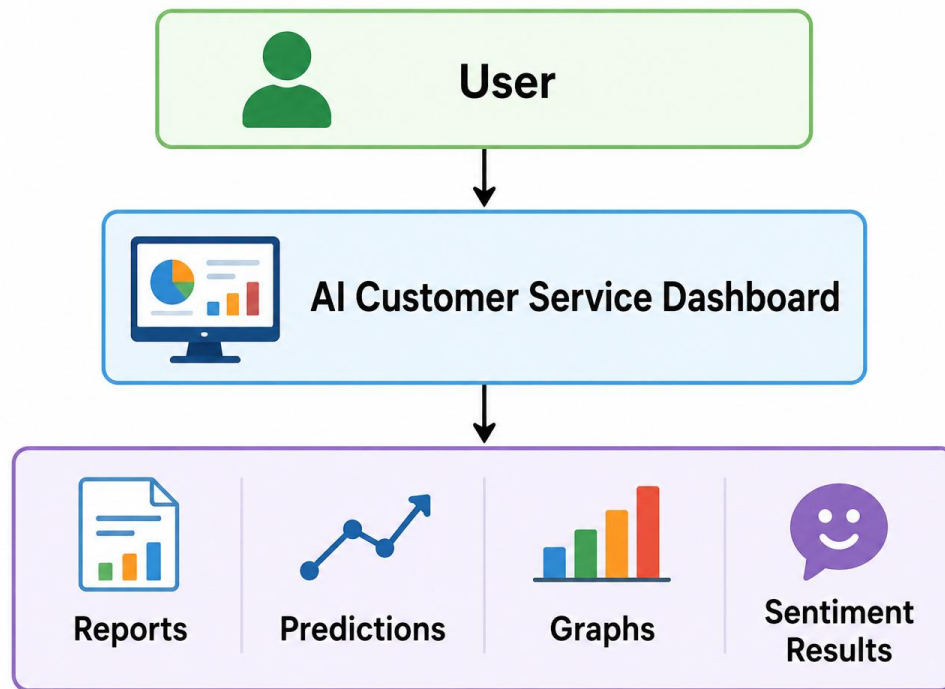


Figure 4.2: Level 0 DFD

4.2.2 Level 1 DFD

The Level 1 DFD does a great job of describing the internal workings of the system. It provides us with a thorough explanation of the Level 1 DFD procedures.

The Level 1 DFD's primary functions are as follows:

- ✚ Information Gathering
- ✚ Storage for databases
- ✚ Information Processing
- ✚ Visualization
- ✚ Predictive Machine Learning
- ✚ Analysis of Sentiment
- ✚ Report Creation

All of these procedures are part of the Level 1 DFD and are necessary for the system to function correctly.

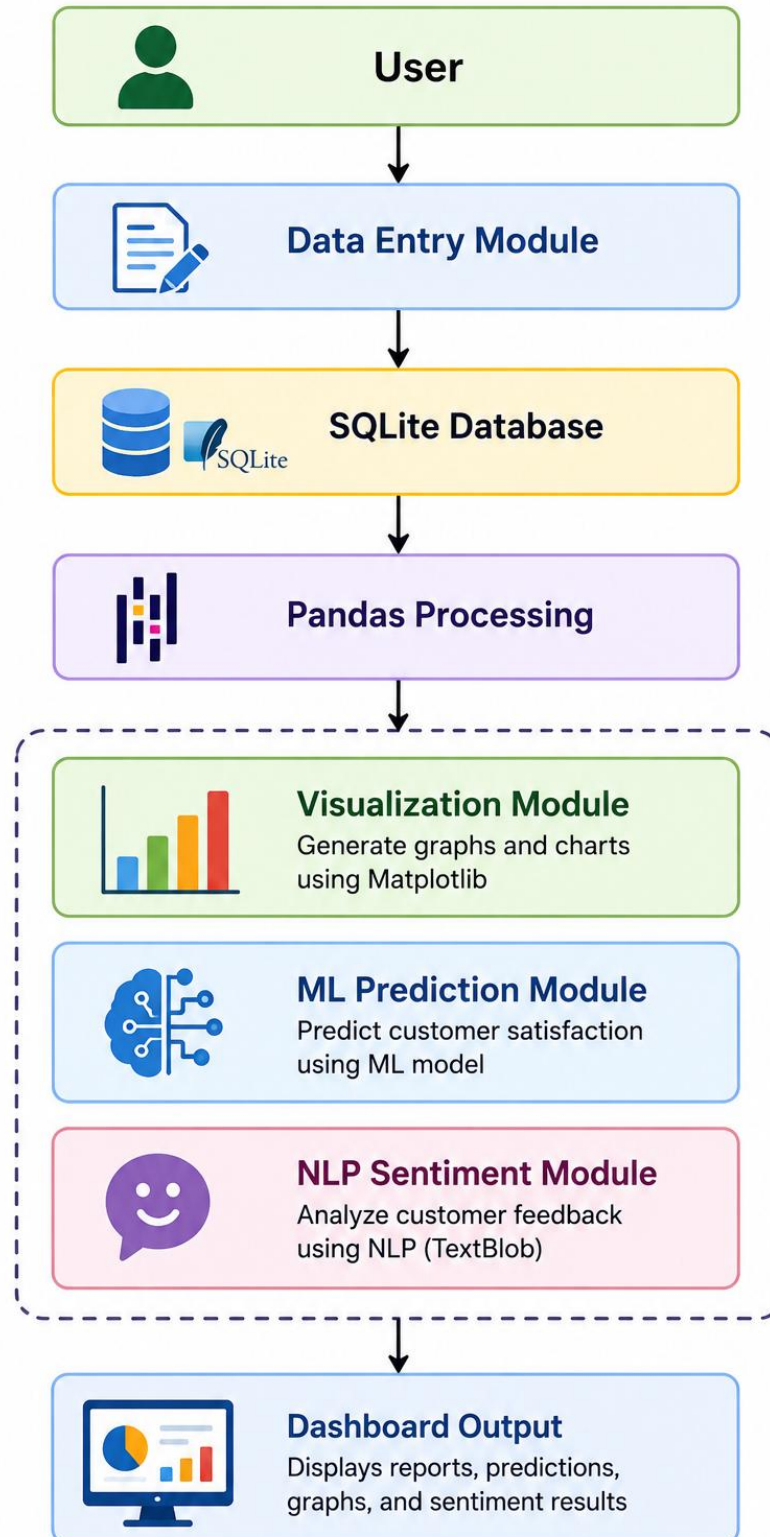


Figure 4.3: Level 1 DFD

4.3 Entity Relationship Diagram (ERD)

Similar to a map, the Entity Relationship Diagram illustrates how the database is configured and how objects are related to one another. Understanding the database greatly depends on the Entity Relationship Diagram.

One table, `customer_data`, is used by the system we are discussing.

All of the client data that customer service representatives need to perform their duties and attempt to predict future events is stored in this `customer_data` table. Information is examined and forecasts are attempted using the `customer_data` table.

Main Entity: customer_data

The entity contains the following attributes:

Attribute Name	Data Type	Description
Company	TEXT	Name of company
AI_Usage_Level	TEXT	Level of AI usage
Customer_Satisfaction	REAL	Satisfaction score
Response_Time	INTEGER	Customer response time

Purpose of ERD

We greatly benefit from the Entity Relationship Diagram.

- It facilitates our comprehension of the database structure.
- The Entity Relationship Diagram allows us to efficiently arrange our data.
- Obtaining the necessary data is also made easy by the Entity Relationship Diagram.
- The Entity Relationship Diagram will help us maintain our system more effectively.

Figure 4.4: ER Diagram



4.4 Database Design

The goal of database design is to organize data in tables for convenient management, storage, and retrieval.

A SQLite database is used in the suggested system because

- It is incredibly light. Not much room is needed.
- It integrates easily and works well with Python.
- One major benefit is that you don't have to set up a server.
- It works well for AI initiatives like this one.

We are utilizing the database file ai_data.db, which contains a table named customer_data.

Table Structure: customer_data

Field Name	Data Type	Description
Company	TEXT	Stores company name
AI_Usage_Level	TEXT	Stores AI usage level
Customer_Satisfaction	REAL	Stores satisfaction score
Response_Time	INTEGER	Stores response time

Database Operations

We use the database for several purposes. The database is used for the following tasks:

1. Insert Operation

This is how we update the database with customer service records. We add customer service entries to the database in this manner. To add customer service records to the database, for instance, we can use this code:

```
cursor.execute("""  
  
INSERT INTO customer_data VALUES (?, ?, ?, ?)  
  
""", (company, usage, satisfaction, response))
```

2. Read Operation

To retrieve data from the database, we utilize the Read Operation. We obtain data from the database in this manner. This code, for instance, can be used to retrieve data from the database:

```
df = pd.read_sql("SELECT * FROM customer_data", conn)
```

3. Filter Operation

The Filter Operation is used to filter customer service records based on the customer service records usage level. We use the Filter Operation to filter customer service records based on the AI usage level of the customer service records. This helps us to look at the customer service records that have an AI usage level.

4. Data Analysis Operation

This device is used to create reports and calculate averages. For this type of activity, we employ Data Analysis Operations. Making sense of numbers so we can understand what is happening is the primary responsibility of the data analysis operation. We can compute averages and produce reports with the use of data analysis operations.

Advantages of SQLite Database

- Fast and lightweight
- Portable database file
- Easy backup and maintenance
- No external server required
- Beginner-friendly

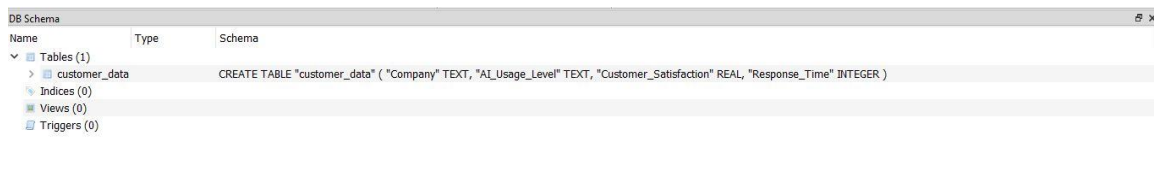


Figure 4.5: SQLite Database Table structure

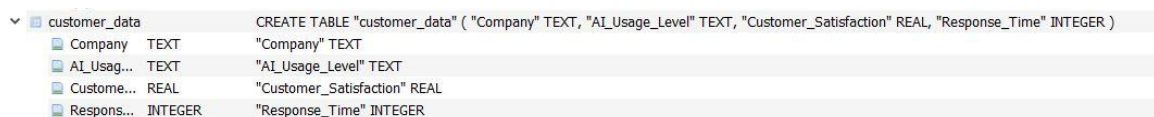


Figure 4.5: SQLite Database customer_data table

4.5 User Interface Design

The application looks good and is simple to use because of its user interface design.

Streamlit is used to create the interface. It contains a few significant items.

- ✓ The user interface design is basic because it is meant to be simple.
- ✓ Additionally, it should be simple to read. It is simple to read the UI design.
- ✓ Because of its engaging design, the user interface is enjoyable to use.
- ✓ Additionally, it is simple to obtain, making the UI design straightforward to use.

The dashboard's components improve the usability of the user interface design.

Main UI Components

A. Title Section

Displays:

- Title of the project
- The header of the dashboard
- Overview of the application

Example:

```
st.title("AI Customer Service Dashboard")
```

B. Sidebar Section

There are numerous items in the sidebar. There are filters in it. Data entry forms are also included. The option to download the report is available.

There are many options available to users. They are able to filter the degree of AI usage. They are able to add records. Reports are also available for download.

C. Dashboard Tab

Displays:

- Average level of satisfaction
- Time of response
- Total number of records
- Preview table of data

This section provides us with a quick overview of the customer service's performance. It displays information such as total records, average response time, and satisfaction. This section focuses on customer service performance. By displaying these elements, it aids in our comprehension of customer service performance.

D. Analysis Tab

We can see some information on this tab. It consists of four items:

1. An illustration of the application of artificial intelligence
2. A histogram displaying people's level of satisfaction
3. A plot of scatter
4. The outcomes of predictions made using machine learning

These graphs are created using Matplotlib.

E. NLP Insights Tab

This section does the following:

1. It checks if text is positive or negative.
2. It figures out if text is good or bad.
3. It shows recommendations.

You can type in what a customer said and find out their feelings away.

User Interface Features

The dashboard provides:

-  Interactive control
-  Real-time filtering
-  Responsive layout
-  Easy navigation
-  Graphical representation of data
-  Simple prediction interface

4.6 Conclusion

The AI Customer Service Dashboard's design is covered in this chapter. The system architecture, entity connection diagrams, data flow diagrams, database configuration, and user interface design are all covered.

Natural language processing, machine learning, analytics, data storage, and visualization are all together in one location with the AI Customer Service Dashboard. This design works well, is scalable, and is simple to use. It aids IT firms in understanding how customer service is impacted by artificial intelligence. Here, the AI Customer Service Dashboard is crucial. It really helps in evaluating the impact of intelligence in customer service.

The AI Customer Service Dashboard's design guarantees that it is user-friendly and capable of handling large amounts of data. The AI Customer Service Dashboard is long-lasting. Provide precise outcomes.

Chapter 5: System Implementation

5.1 Tools and Technologies Used

When it comes to software development, the implementation stage is crucial. At this point, the design we have in mind is transformed into a functional system. In order to create an AI-based customer service dashboard for this project, we are utilizing a variety of technologies and Python packages. This dashboard is intended for IT firms. Our goal is to create a system that can analyze data, create graphs and charts, make predictions using machine learning, and use sentiment analysis to determine people's emotions.

Because Python is so user-friendly and versatile, we decided to use it for this project. Libraries for data science and artificial intelligence are also well supported by Python. Python is used in the development of an AI-based customer service dashboard. Python is an option for applications that rely on artificial intelligence. Because Python is open-source and has a user base, many people utilize it for artificial intelligence. For this reason, we employ Python for our AI-based customer service dashboard..

To complete tasks, the project makes use of a few tools and technologies.

5.1.1 The Python Programming Language

For this project, Python is the language of choice. Python has words and is easy to read, making it simple to comprehend and utilize. Python excels in the following areas:

1. Examining data
2. The use of machines
3. Developing online apps
4. Language comprehension
5. Making database connections

Because Python excels at all of these tasks, we utilize it for them. Machine learning and data analysis are made simple with Python. Additionally, we employ Python for natural language processing and online application development. Python is excellent for connecting to databases.

Python is used in the project for the following purposes:

- Establishing a SQLite database connection
- Handling client information
- Making visualizations
- Developing models for machine learning
- Using Python for sentiment analysis

Advantages of Python

- Learning and using Python for projects is simple.
- There are a lot of libraries available for Python.
- Python is compatible with all platforms.
- Python is available as open-source software.
- Python is useful for machine learning and artificial intelligence applications.

5.1.2 Streamlit Framework

Our system's webpage is created using Streamlit. It is a technology that enables us to create interactive web pages.

Streamlit is used extensively in this project. Our website's layout is created using Streamlit. Additionally, streamlit is used to create user-friendly filters. Charts and graphs are displayed using streamlit. Streamlit is used to collect user data. Streamlit is used to illustrate our expectations. Streamlit is used to convey people's emotions.

The following are some of Streamlit's advantages:

- It is simple to develop code for Streamlit;
- We can create things rapidly with Streamlit;
- It has a lot of useful features;
- It updates on time;
- It integrates with other Python tools.

There are many components in our Streamlit website. There are controls on the side of our Streamlit website. Data uploading and filtering options are available on our Streamlit website. There are panels on our Streamlit website that display data images. There is a section on our Streamlit website where you may view user comments. There is an area on our Streamlit website where you can make predictions.

The application is executed using the command:

```
streamlit run app.py
```

This command launches the application in a web browser.

5.1.3 SQLite Database

Customer service data is stored in SQLite. All of the records are kept here. All of this data is contained in a file named:

ai_data.db

SQLite is easy to use and doesn't require a server. Projects and medium-sized applications are ideal for it. There are some significant items in the database:

1. The company's name
2. The extent to which the business employs intelligence
3. The degree of consumer satisfaction
4. The time it takes to reply to clients

All of this data is stored in the SQLite database, which is an excellent tool for handling customer service records similar to those found in the *ai_data.db* file.

Benefits of SQLite

- ❖ SQLite's tiny size is one of its features.
- ❖ It moves really quickly.
- ❖ SQLite is quite easy to set up.
- ❖ SQLite is quite convenient because it doesn't require a server.
- ❖ SQLite is incredibly resource-efficient and works well for programs that run on your computer.

The project connects SQLite with Python using the `sqlite3` module.

Example:

```
conn = sqlite3.connect("ai_data.db")
```

The database table used:

customer_data

5.1.4 Pandas Library

When working with data, the Pandas library is really helpful. Python's Pandas package is used for a variety of data-related tasks. The Pandas library is utilized for several tasks in this project. Using the Pandas library, one can:

- Obtain information from the SQLite database.
- Examine the necessary records.
- Check out what our clients are doing.
- Determine a few key figures.
- Display tables on the dashboard

DataFrames are a type of data storage used by the Pandas library. This facilitates the analysis of the data and its interpretation. DataFrames from the Pandas library are quite useful.

Example:

```
df = pd.read_sql("SELECT * FROM customer_data", conn)
```

Pandas-based functions:

- To filter data, we utilize Pandas.
- It facilitates computation.
- Pandas is used for data transformation.
- We use Pandas to manage records.

Benefits

- Pandas provides processing.
- It facilitates data handling.
- Pandas is capable of analysis.

5.1.5 Matplotlib Library

Matplotlib is an excellent tool for displaying data. The system produces:

- Bar graphs
- Histograms
- Plots of scatter

These graphs help people see things like:

- The degree of client satisfaction
- The frequency of AI use
- How customer satisfaction is impacted by response time

For instance:

```
fig, ax = plt.subplots()

df['Customer_Satisfaction'].hist(ax=ax)
```

The advantages of visualization are very beneficial:

1. You are able to interpret things.
2. Because you have a clearer understanding of the situation, you can make decisions.
3. Trend analysis benefits from visualization as well.
4. It is helpful for tracking the visualization's and its tools' performance.

The primary benefit of visualization is that it facilitates improved decision-making and performance monitoring, among other things.

5.1.6 Scikit-learn Library

Machine learning techniques are implemented with Scikit-learn. For machine learning, Scikit-learn is quite useful. Linear regression is used in the project. We utilize Scikit-learn to determine how satisfied consumers are with our service depending on how long it takes us to reply to them. Our goal is to forecast customer happiness based on response time using Scikit-learn. In this manner, we may estimate customer happiness based on response time using Scikit-learn.

Example:

```
model = LinearRegression()  
  
model.fit(X, y)
```

Machine learning is really helpful because it allows us to do many things. It helps us predict what might happen in the future and understand what our customers are doing. Machine learning is great for making decisions.

The good things about machine learning are

- It is easy to set up and start using machine learning
- Machine learning gives us accurate results
- The prediction models we get from machine learning are reliable.

5.1.7 TextBlob Library

Natural language processing makes use of TextBlob. This implies that it attempts to ascertain people's emotions by listening to what they have to say. It examines client comments. determines the customer's level of happiness, sadness, or contentment.

It can determine if a consumer feels positively, negatively, or neutrally about a good or service. We can better comprehend the thoughts and emotions of our customers thanks to TextBlob.

For instance:

```
blob = TextBlob(text_input)  
  
score = blob.sentiment.polarity
```

Natural language processing is crucial because it makes things easier for us.

- We are able to comprehend how clients are feeling.
- We can quickly review a large amount of feedback.
- With our goods and services, we can increase consumer satisfaction.

5.1.8 Visual Studio Code

Visual Studio Code, generally known as VS Code. Among Visual Studio Code's features are:

1. A coding editor that facilitates my writing of code
2. Debugging support, which allows me to identify errors in my code
3. A Visual Studio Code terminal
4. Add-ons for Python, the programming language I utilize

The following are some benefits of utilizing Visual Studio Code:

- Writing code in Visual Studio Code is simple.
- Visual Studio Code is use to debug code more quickly.
- Visual Studio Code facilitates better project management for me.

5.2 System Development Methodology

The project is carried out in a structured manner. This means that everyone involved in the project follows a specific process to ensure everything is done right. They start by planning what the software should do, then they build it, check it to make sure it works, and finally they use it. The project always goes through this sequence to ensure the software is developed properly.

The methodology used in this project follows these phases:

1. Requirement Analysis
2. System Design
3. Database Design
4. Coding and Development
5. Testing
6. Deployment
7. Maintenance

5.2.1 Requirement Analysis

In this part of the project, we figured out what the system needed to do.

The system had to do several things. It needed to:

- Keep track of customer service information
- Check how happy customers were
- Create reports that are easy to understand
- Try to guess how happy customers will be in the future
- Find out what customers really think

We made a list of what the system had to be able to do. This includes:

- ✓ Adding new customer information
- ✓ Sorting through data
- ✓ Making charts
- ✓ Trying to predict things
- ✓ Understanding what customers are saying

We also thought about what the system should be like. It should be:

- Easy for people to use
- Work quickly
- Be accurate
- Be able to handle a lot of work

5.2.2 System Design Phase

In this part of the project, we determined what the system would look like and how it would function. To better understand these aspects, we created some diagrams.

The diagrams we developed include:

- System Architecture Diagram
- Workflow Diagram
- Data Flow Diagram (DFD)
- Entity-Relationship (ER) Diagram

We divided the system into sections called modules. The system includes the following modules:

- ✓ Database Module
- ✓ Visualization Module
- ✓ Machine Learning Module
- ✓ Natural Language Processing Module

5.2.3 Database Design Phase

The SQLite structure was created. It includes a table named customer_data. This table has several fields: Company, AI_Usage_Level, Customer_Satisfaction, and Response_Time. The SQLite database structure is efficient because it handles tasks well.

For example:

- It makes it simple to retrieve the data we need
- It stores data in a way that uses less space
- It allows quick processing of information

The customer_data table and the SQLite database structure are designed to work together smoothly. The customer_data table and the entire SQLite database are set up to make our work easier.

5.2.4 Coding and Development Phase

We completed the project in this section.

Here's what we did:

- We created the user interface using Streamlit.
- We linked the database.
- We included the graphs.
- We integrated Machine Learning into the project.
- We added Natural Language Processing to the project.

The code was written in a straightforward way, making it easier to fix any issues that may arise with the Machine Learning and Natural Language Processing parts.

5.2.5 Testing Phase

The system was checked to ensure it works properly.

- It needed to produce the correct output.
- It had to run without any mistakes.
- It had to display information in a clear way.
- It had to make predictions.

There are different types of testing done for the system. These tests include:

- ✓ The system is tested to make sure it works as expected. This is called functional testing.
- ✓ The system is tested with different inputs. This is called input testing.
- ✓ The system is tested to check if its predictions are accurate. This is called prediction testing.
- ✓ The system is tested to ensure the database functions correctly. This is called database testing.

5.2.6 Deployment Phase

The application was executed locally using Streamlit.

Execution command:

```
streamlit run app.py
```

The dashboard opens in a web browser.

5.2.7 Maintenance Phase

The system can be updated in the future by doing a things.

1. We can add Artificial Intelligence models to the system.
2. The system can also be updated by improving the Natural Language Processing analysis that the system uses.
3. There are an other ways to update the system, such, as connecting to cloud databases and enhancing the visualization that the system provides.
4. The system can be updated by enhancing the visualization of the Artificial Intelligence models and the Natural Language Processing analysis.

5.3 Module Description

The project is divided into different modules.

5.3.1 Database Module

The Database Module is used to store customer service data. This is where we keep all the information, about our customers. The Database Module does a things. It can insert records retrieve records and manage customer information. We use SQLite and sqlite3 to make this work.

5.3.2 Dashboard Module

The Dashboard Module provides a user interface. This is what the users see when they interact with our system. The Dashboard Module has a functions. It can display metrics show tables and manage filters. We use Streamlit to make the Dashboard Module work.

5.3.3 Visualization Module

The Visualization Module is used to display graphs and charts. This helps us understand the data better. We use a types of graphs. We use bar graphs, histograms and scatter plots. The Visualization Module uses Matplotlib to make these graphs.

5.3.4 Machine Learning Module

The Machine Learning Module is used to predict customer satisfaction. This means it tries to figure out how happy our customers are. The Machine Learning Module uses an algorithm called Linear Regression. It looks at the response time. Then it predicts the satisfaction score of the customer.

5.3.5 NLP Module

The NLP Module is used to analyze customer feedback. This means it looks at what our customers are saying about us. The NLP Module uses a technology called TextBlob. It can tell us if the customer feedback is positive, negative or neutral.

5.3.6 Report Generation Module

The purpose of the Report Generation Module is to generate reports that people can download. The Report Generation Module contains the following things:

- ✓ Satisfaction
- ✓ Total records
- ✓ Insights, about the Report Generation Module

5.4 AI Model Integration (Chatbots, NLP, etc.)

Artificial Intelligence plays an important role in this project.

The project covers a few key areas: Machine Learning, Natural Language Processing, and Predictive Analytics.

5.4.1 Machine Learning Integration

Machine Learning uses a method called Linear Regression. Its main purpose is to determine customer satisfaction.

Here's how it works:

- Load the data
- Train the system
- Get input from the user
- Provide a prediction

For instance, you can create a model like this:

```
model = LinearRegression()
```

```
model.fit(X, y)
```

The benefits of using Machine Learning include:

- It can make predictions
- It helps in making data-driven decisions
- It simplifies the process of evaluating performance

This way, we can benefit from Machine Learning.

5.4.2 NLP Integration

TextBlob is used to implement NLP. Checking what clients are saying is the primary objective.

This is how it operates:

- A user composes their comments.
- TextBlob then examines the feedback's polarity to determine if it is positive or negative.
- After that, our system determines whether the feedback is neutral, negative, or favorable.

The possible outcomes are: Positive, Negative or Neutral

For example:

```
blob = TextBlob(text_input)
```

```
score = blob.sentiment.polarity
```

5.4.3 Benefits of AI Integration

AI integration has benefits.

- One of the benefits is that AI helps with faster analysis.
- This leads to an understanding of customers.
- AI also enables decision-making.
- Overall AI integration results in efficiency.

5.5 Conclusion

The implementation phase worked well. It turned the proposed AI Customer Service Dashboard into a working app. The project brings together database management, visualization, machine learning and NLP. This is all done in one platform using Python and Streamlit.

The system shows how Artificial Intelligence can make customer service analysis better. It does this through automation, prediction, visualization and sentiment analysis. The AI Customer Service Dashboard also provides a base. This base can be improved in the future with things like chatbot integration. Cloud deployment and advanced AI models are also possible.

We can also improve it more with chatbot integration. Move it to the cloud. The AI Customer Service Dashboard has a foundation, for these future enhancements.

Chapter 6: Testing

6.1 Introduction to Testing

Testing is an important part of making software. It makes sure that the system works the way it is supposed to. Testing helps find mistakes makes the system work better and checks that all parts of the system are working together like they should.

For this project we tested the AI Customer Service Dashboard. This dashboard was built using Python, Streamlit, SQLite, Machine Learning and Natural Language Processing. We tested it to make sure it can do its job. The AI Customer Service Dashboard is supposed to look at data make charts make predictions and figure out how people are feeling about things.

When we tested the AI Customer Service Dashboard we checked a things.

1. We checked that the dashboard can connect to the database.
2. We checked that the dashboard works like it is supposed to.
3. We checked that the dashboard can make graphs.
4. We checked that the Machine Learning part of the dashboard can make predictions.
5. We checked that the sentiment analysis part of the dashboard can figure out how people are feeling.
6. We checked that the user interface works well and is easy to use.

We tried the AI Customer Service Dashboard with inputs and scenarios to make sure it works well and gives accurate results. The AI Customer Service Dashboard is a part of this project and testing it was very important.

6.2 Objectives of Testing

The main objectives of testing are:

- To make sure the system works correctly
- To Fix errors
- To check if database operations are working fine
- To see if prediction accuracy is good
- To test sentiment analysis
- To make the system perform better and be reliable

Testing helps ensure that the application we developed meets the requirements for evaluating how Artificial Intelligence affects customer service for tech companies. The testing process helps ensure that Artificial Intelligence in customer service works properly for tech companies. We test to ensure Artificial Intelligence functions in customer service. Testing also helps to validate that Artificial Intelligence prediction accuracy is good, in customer service.

6.3 Types of Testing Performed

We used a variety of testing methods when we were working on the project. The testing methods we used include:

- Unit Testing
- Integration Testing
- Functional Testing
- Database Testing
- Machine Learning Testing
- NLP Testing
- User Interface Testing

We did all these types of testing to make sure the project was good. The project had Unit Testing and Integration Testing and Functional Testing. We also did Database Testing and Machine Learning Testing and NLP Testing. Additionally we did User Interface Testing to see how the project looked to the users.

6.3.1 Unit Testing

Unit testing is a way to check parts of a program to see if they work correctly. The following parts of the program were checked:

- The Database loading module
- The Data filtering module
- The Prediction module
- The Sentiment analysis module

Example:

The `load_data()` function was tested to ensure that customer data loads correctly from the SQLite database.

Expected Result: Data should load successfully.

Actual Result: Data loaded correctly.

Status: Passed

6.3.2 Integration Testing

Integration testing makes sure different parts work together.

The following integrations were tested:

- Streamlit with SQLite database
- Machine Learning model with dashboard
- NLP module with user interface
- Visualization module with processed data

The testing showed that all parts, like Streamlit SQLite database, Machine Learning model, dashboard, NLP module, user interface, Visualization module and processed data worked correctly and gave results. All modules, including Streamlit Machine Learning model, NLP module and Visualization module communicated correctly.

Status:Passed

6.3.3 Functional Testing

The project requirements are checked to make sure that the features of the project work as they should.

The things that are tested include:

1. Loading records
2. Adding new data
3. Filtering data
4. Generating graphs
5. Predicting satisfaction
6. Performing sentiment analysis
7. Downloading reports

The testing is done to verify that the project works as it should so the features of the project are checked one by one to make sure that the features of the project work according to the project requirements. The features of the project that are tested include the features of the project, like loading records and adding data and other features of the project.

For Example, when the user chooses the AI usage level as High, the system should show the records that have High AI usage. This is what we expect to happen. In this case the records were filtered correctly. So the status of this test is Passed.

The records, with High AI usage were displayed as they should be. This means the test was successful and the result is that it Passed.

6.4 Test Cases

The following test cases were performed during testing.

Test Case ID	Test Description	Expected Result	Status
TC01	Load database	Data loads successfully	Passed
TC02	Add new record	Record inserted successfully	Passed
TC03	Filter data	Correct records displayed	Passed
TC04	Generate graphs	Charts displayed properly	Passed
TC05	Predict satisfaction	Prediction generated	Passed
TC06	Analyze sentiment	Correct sentiment shown	Passed
TC07	Download report	Report downloaded successfully	Passed

6.5 Error Handling

The system has error handling to stop the application from failing. The project deals with these problems:

- ✓ When the database is missing
- ✓ When the user types something
- ✓ When there are mistakes in the predictions
- ✓ When there is no text, in the input

Example:

try:

```
conn = sqlite3.connect("ai_data.db")
```

except:

```
st.error("Database not found")
```

This improves the stability and reliability of the system.

6.6 Advantages of Testing

The testing process had some good outcomes:

- ❖ It made the system more reliable
- ❖ It helped with getting accurate predictions
- ❖ It made sure database operations had no errors
- ❖ Users had an experience
- ❖ The system generated visualizations
- ❖ The sentiment analysis was accurate

Testing made sure the system worked well in various situations.

Chapter 7: Results & Discussion

7.1 Introduction

This chapter is about the results we got from the AI Customer Service Dashboard. It talks about how the system is doing overall. We made this project to see how Artificial Intelligence is doing in customer service for tech companies. We looked at things like how happy customers are, how long it takes to respond, how much Artificial Intelligence is being used, and what customers are saying about it.

The system puts together a lot of things like managing databases, making data look nice, machine learning, and understanding language all in one place. The results from the dashboard help us understand how Artificial Intelligence technologies make customer service better and faster for customers.

The chapter also talks about what the graphs mean, what the predictions say, what the sentiment analysis results show, and the good and bad things about the system we developed.

7.2 Results Obtained from the System

The Customer Service Dashboard made some helpful things from the customer service information that is kept in the SQLite database.

The main things the system came up with are:

- ❖ Customer satisfaction analysis
- ❖ Response time analysis
- ❖ AI usage distribution
- ❖ Machine learning prediction
- ❖ Sentiment analysis results
- ❖ Visualization reports

The Customer Service Dashboard results help the Customer Service Dashboard organizations see how the customer service is doing and make choices about the customer service. The customer service results from the Customer Service Dashboard are very useful, for the customer service.

7.3 Customer Satisfaction Analysis

The system is really good at figuring out how happy customers are. It looks at all the information it has. Calculates the average score of how satisfied customers are.

The system shows us a few things:

- How satisfied customers are on average

- If customer satisfaction is going up or down over time
- How the time it takes to respond to customers affects how satisfied they are

From looking at all the information we found out that:

- ✚ When companies respond to customers quickly the customers are usually happier.
- ✚ Companies that use Artificial Intelligence a lot tend to be better at helping their customers.
- ✚ When companies take long to respond customers are not as happy.

We used a graph made by Matplotlib to see how customer satisfaction scores are spread out.

This analysis shows that Artificial Intelligence can really help make customers happier when it is used in a way. The customer satisfaction analysis is a part of what the system does. Artificial Intelligence has an impact, on customer experience.

7.4 AI Usage Distribution Analysis

The dashboard looks at how different companies use Artificial Intelligence when they deal with customers. The way companies use Artificial Intelligence is divided into three groups: High, Medium and Low. A bar graph was made to show how many companies fall into each group.

The results showed that:

- Companies that use a lot of Artificial Intelligence can handle customer interactions efficiently.
- Artificial Intelligence automation helps reduce the workload, on human customer service agents.
- Companies that use Medium and Low levels of Artificial Intelligence take longer to respond to customers.

This analysis shows that using Artificial Intelligence can make customer service work better. Artificial Intelligence helps companies handle customer service in an efficient way. The use of Artificial Intelligence can really improve how customer service systems work.

7.5 Response Time Analysis

Response time is really important when it comes to how happy customers. The system looked at response time in a few ways:

- ✓ It made scatter plots
- ✓ It looked at the average time it took to respond
- ✓ It used machine learning to make predictions about response time

When you look at the scatter plot it shows how response time and customer satisfaction are connected.

The results were pretty clear:

- When response time is shorter people usually give feedback.
- When response time is long it makes customers unhappy.
- Using intelligence systems really helps cut down on waiting time.

You can see it on the dashboard. Systems that use intelligence are faster and more efficient than the old way of doing customer service. The artificial intelligence systems are just better, at getting things done quickly.

7.6 Machine Learning Prediction Results

The project uses a Linear Regression model to figure out how happy customers are, with the service they get based on how it takes to respond to them. We used information from customer service to teach the machine learning model. This information is stored in the SQLite database.

The part of the project that makes predictions worked well it

- Took in what the user had to say
- Looked at the data on how long it took to respond
- Came up with scores that showed how happy customers would be

For example:

- ❖ When we responded quickly the customers were really happy
- ❖ When we took a time to respond the customers were not very happy

What we found out shows that machine learning can really help companies:

- ✚ Know what customers are going to do
- ✚ Make their service better
- ✚ Make decisions based on facts and numbers

Even though the model we used is not very complicated it shows in a simple way how Artificial Intelligence can be used in a practical way to understand what customers want when it comes to service.

7.7 Conclusion

The results, from the AI Customer Service Dashboard show that Artificial Intelligence really helps tech companies do better at customer service.

The system we developed does some things:

- It keeps track of customer service data
- It analyzes data in real time
- It makes visual reports
- It guesses how satisfied customers will be
- It looks at customer feedback using NLP

This project shows how Artificial Intelligence, Machine Learning and Natural Language Processing can be used in real-life customer service systems.

Overall our system met its goals. Proved that AI technologies can help companies work more efficiently save money and make customers happier. Artificial Intelligence and AI technologies make a difference.

Chapter8: Conclusion & Future Scope

8.1 Introduction

Artificial Intelligence is really important for customer service these days. Lots of organizations are using Artificial Intelligence to make customer service better. They want to talk to customers easily answer questions faster and do things automatically that they do over and over again. This makes the service they provide a lot better.

This project is called "Evaluating the Impact of Artificial Intelligence in Customer Service for Tech Companies". We worked on making a dashboard for customer service that uses Artificial Intelligence. We used some tools like Streamlit, SQLite, Pandas, Matplotlib, Machine Learning and Natural Language Processing to build it.

Our system looks at customer service information shows us numbers and charts predicts if customers are happy and figures out what customers mean when they give feedback. We showed that Artificial Intelligence can really help make customer service better and faster. It can also help people make decisions.

This part of the project sums up what we found out and what we think should happen next. We also have some ideas for how to make the project better in the future. Artificial Intelligence is a part of this and we think it can do a lot for customer service. The project is, about Artificial Intelligence and customer service and how Artificial Intelligence can make customer service better for Tech Companies.

8.2 Conclusion

The new AI Customer Service Dashboard did a job of meeting the main goals of the project. This system gives us a place to look at how well the customer service is doing using Artificial Intelligence methods.

The application lets users do these things:

- Put and manage customer service information
- Look at customer satisfaction and how long it takes to respond
- Sort records by how much Artificial Intelligence is used
- Guess how happy customers will be using Machine Learning

- Look at what customers are saying using Natural Language Processing
- Make reports that tell us a lot

Having all these things in one dashboard makes the system work well and is easy to use.

One of the things we did well in this project is making Machine Learning and Natural Language Processing work in a real customer service application. The Linear Regression model showed us how the time it takes to respond can affect how happy customers are. The TextBlob library was also good at figuring out if customers were happy, sad or okay, with the service. The AI Customer Service Dashboard and Artificial Intelligence really helped us with this.

The project showed how important it is to use data visualization to understand what customers do and how well services work. Using Matplotlib to make graphs and charts made it easier for people to look at the information and understand it.

The SQLite database was a way to store customer information. It helped keep all the data consistent and made it possible to get to the information away through the dashboard.

Streamlit made the user interface a lot better by making it interactive and easy to use. People can easily move around pick what they want to see add information and look at the data without needing to know a lot, about technology.

From looking at everything we saw that Artificial Intelligence can really help customer service systems in ways.

- It can make things happen faster
- It can make things work better
- It can make customers happier
- It can look at information and make reports automatically
- It can reduce the amount of work people have to do
- It can help people make decisions

The project also found some problems with customer service systems that use Artificial Intelligence. Artificial Intelligence models can sometimes fail to understand what the customer is asking and how well they can predict things depends a lot on the quality of the

data they have. Also simple Natural Language Processing models may not always understand how the customer is feeling or what they really mean.

With these problems the project showed that Artificial Intelligence can really help tech companies do a better job with customer service.

Overall the system we developed shows that Artificial Intelligence technologies can be used with tools to see how well customer service is doing. This is a way to use Artificial Intelligence and other tools, like analytics and visualization to evaluate customer service performance.

8.3 Future Scope

The future of Artificial Intelligence in customer service is really big. It is getting bigger all the time. Artificial Intelligence technologies are going to get more advanced and smarter in the years to come. This project can be made better in a lot of ways.

Advanced Chatbot Integration

We can make future versions of the system have chatbots that are powered by Artificial Intelligence and can talk to customers in time on their own.

Deep Learning Implementation

We can use learning to make the system better at predicting things and to understand how customers behave in a smarter way.

Real-Time Customer Monitoring

The dashboard can be connected to customer support systems so we can see what customers are doing in time.

Voice-Based Sentiment Analysis

Future systems can listen to what customers are saying and figure out how they are feeling by using special technologies that can recognize speech.

Cloud Deployment

We can put the application on the cloud with companies, like Amazon Web Services or Azure or Google Cloud more people can use it and it can handle more users.

Big Data Integration

We can look at big sets of customer service data using Big Data technologies to get more ideas and do more advanced analysis of Artificial Intelligence and customer service.

AI Recommendation Systems

The system can tell people what they might like based on what they have done and what they like. It looks at what the customer does and what they have done in the past to give them ideas or suggestions.

Mobile Application Development

We can make a version of the dashboard that works on phones. This way people can get to it and check on things easily.

Advanced Visualization Techniques

We can use tools like Plotly or Power BI to make really cool and interactive pictures that help people understand things better.

Enterprise-Level Integration

The system can work with tools that big companies use, like the tools they use to manage customers and help them with problems.

The future of AI in customer service is going to be about making things automatic, personal and smart. AI is going to help companies understand what customers want and make them happy. As AI gets better and better companies will use it more and more to make customers happy and to make their businesses run smoothly. The future of AI, in customer service will focus on AI. AI will make customer service better. AI will help companies do their jobs.

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Appendices

Source Code

The project has a lot of code. This code includes

- The code for the Streamlit dashboard
- The part where we connect to the SQLite database
- The Machine Learning model that we use
- The way we do Sentiment analysis
- and the modules, for Visualization and reporting.

Main files that we use are:

- ❖ run.py
- ❖ create_db.py
- ❖ app.py
- ❖ requirements.txt

We need these files to make the project work. The project uses Streamlit dashboard code and SQLite database connectivity and Machine Learning model integration and Sentiment analysis implementation and Visualization and reporting modules.

Example:

```
import streamlit as st

import pandas as pd

import sqlite3

from sklearn.linear_model import LinearRegression

from textblob import TextBlob
```

The source code shows us how Artificial Intelligence is used in the customer service dashboard. This means Artificial Intelligence techniques are really a part of the customer service dashboard. We can see how Artificial Intelligence helps people when they need assistance, from the customer service team.

Database Details

Database Name: ai_data.db

Table Name: customer_data

Fields Used:

Field Name	Data Type	Description
Company	TEXT	Company name
AI_Usage_Level	TEXT	AI usage category
Customer_Satisfaction	REAL	Satisfaction score
Response_Time	INTEGER	Response time

Sample Database Query:

```
SELECT * FROM customer_data;
```

The SQLite database is where we keep customer service information. We use this information to look at things and make predictions about the future. The customer service information in the SQLite database is really important, for this.

User Manual

Steps to Run the Project

Step 1: Install Anaconda

Install Anaconda from:

[Anaconda Official Website](#)

Anaconda includes:

- ❖ A Python environment
- ❖ A Conda package manager
- ❖ Support for Jupyter Notebook
- ❖ Pre-installed data science libraries

Step 2: Open Anaconda Prompt

After you install Anaconda you should do a couple of things. First open the Anaconda Prompt.

Next you need to go to the project folder.

To do this use the following command:

```
cd path_to_project_folder.
```

You will be using Anaconda to work on your project so make sure you are, in the folder. The Anaconda Prompt is where you will be typing the commands to navigate to the project folder.

Example:

```
cd C:\Users\Dev\Documents\AI_Project
```

Step 3: Install Required Libraries

To get everything we need we have to do a couple of things. First we can try running the following command:

```
> python run.py
```

This will install all the required libraries. for the project to run.

We can do it another way. We can install the required libraries by running the following command:

```
> pip install streamlit pandas matplotlib scikit-learn textblob
```

This way we can be sure that we have streamlit and other libraries like pandas and matplotlib and scikit-learn and textblob.

Step 4: Create Database

To create a database, we will run the command:

```
python create_db.py
```

This command creates a database file named: ai_data.db

Step 5: Run the Streamlit Application

To run your Streamlit application follow these steps:

Type the command:

```
> streamlit run app.py
```

This will start your Streamlit application. You can now view it in your web browser.

Step 6: Open the Dashboard

After running the command, Streamlit automatically opens the dashboard in the browser.

The dashboard includes:

- Customer service analytics
- Graphs and charts
- Machine learning prediction
- Sentiment analysis
- Report generation

Required Files

The project folder should contain:

- run.py
- app.py
- create_db.py
- ai_data.db
- requirements.txt

Technologies Used

- Python
- Streamlit
- SQLite
- Pandas
- Matplotlib
- Scikit-learn
- TextBlob